

Impact of Local Time on march 17th 2015 Geomagnetic Storm response at mid-low latitudes

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Abstract

This study analyzes the impact of the partial ring current and Prompt Penetration Electric Fields (PPEFs) on the local geomagnetic response, using data from four mid-to-low latitude observatories separated by approximately 60°–90° in longitude, covering different local time sectors. We focus on the geomagnetic storm of 17 March 2015. Using Fourier analysis, we reconstruct the magnetic signatures of ionospheric currents. Our results demonstrate that the local geomagnetic response is most significantly affected in the dawn and dusk sectors, where the contribution of the partial ring current to the local magnetic record is most pronounced.

Keywords: Regional Space weather, Magnetic fluctuations, Prompt-Penetration Electric Fields

1 Introduction

A geomagnetic storm (GS) is a temporary disturbance of Earth’s magnetosphere caused by the injection of energy and charged particles into the system, leading to a significant reduction in the geomagnetic field intensity. These GS are typically triggered by solar phenomena such as Coronal Mass Ejections (CMEs) or high-speed solar wind (HSSW) streams. Due to their potential impacts on technological systems—including disruptions to communications, navigation, and power grids—GSs are among the most extensively studied phenomena in space weather research.

During the Main Phase (MP) of a GS, energy and charged particles are transported into Earth’s inner magnetosphere. Since the magnetosphere is dynamically coupled with the ionosphere, particularly at high latitudes [C.T Russell, 2016], the ionosphere exhibits pronounced responses to these perturbations. One critical ionospheric effect is the generation of storm-time electric fields, which can be manifested at different latitudes:

- Prompt Penetration Electric Fields (PPEF) [Nishida, 1968a,b]
- Disturbed Dynamo Electric Fields (DDEF) [Blanc and Richmond, 1980]

The concept of PPEF was first introduced by Nishida [1968a,b], who observed periodic magnetic fluctuations between 4 hours and 30 minutes in ground magnetometers located around the auroral oval and the dip equator, both showing nearly identical fluctuations. Later, Vasyliunas [1970], Jaggi and Wolf [1973] developed the first model explaining the penetration of convective electric fields ($\mathbf{E}_C$) during disturbed periods, propagating through Region 1 Field-Aligned Currents (R1-FAC). During quiet periods, $\mathbf{E}_C$ is shielded by a polarized electric field induced by Region 2 Field-Aligned Currents (R2-FAC). However, during the MP of a GS, magnetic reconnection leads to an almost immediate intensification of R1-FAC activity due to the injection of solar wind particles and energy. Since R2-FAC depends on ring current activity to strengthen, it initially fails the shielding of $\mathbf{E}_C$, allowing its transfer to the ionosphere through R1-FAC. This process continues until R2-FAC intensifies its activity though the intensification of ring current, allowing R2-FAC balancing R1-FAC activity, which can take from tens of minutes to several hours [Wei et al., 2015]. During this period of imbalance, $\mathbf{E}_C$ penetrates into the mid- and low-latitude

ionosphere, being that the reason to be called Penetration Electric fields. Furthermore, Kelley et al. [1979], Wei et al. [2015] showed that PPEF depends on the southward orientation of the interplanetary magnetic field (IMF) component B_Z and can also be triggered by sharp directional changes in B_Z .

DDEF on the other hand, are associated with an ionospheric current driven by the injection of charged particles into the auroral ionosphere. This injection, produces a Joule heating and leads to the circulation of equatorward thermospheric winds which drag charged particles (protons) Blanc and Richmond [1980]. With the dragged protons, it gets to a charge separation which induces a Pedersen current in the same direction of neutral wind flow. Near the equator, the Coriolis effect produces proton accumulation and westward circulation occur around the dip equator, generating a poleward electric field. Consequently, a second Pedersen current is induced in the poleward direction, opposing the first current. At mid-latitudes ($\sim 40^\circ$), the above commented phenomena, drives Hall currents, which are interrupted at the terminators (limit of day-side with night-side) and get a closure or circuit at adjacent latitudes. This process results in vortex-shaped currents known as the Disturbed Dynamo, characterized by directions opposite to the Solar Quiet (SQ) current. For this reason, the disturbed dynamo is also referred to as the anti-SQ current.

At mid-low latitudes ($20 - 30^\circ$), geomagnetic storms can significantly affect electron density through two primary mechanisms: Prompt Penetration Electric Fields (PPEF) and disturbance dynamo-driven neutral winds. On the dayside, an eastward PPEF causes an upward $\mathbf{E} \times \mathbf{B}$ drift, lifting the plasma to higher altitudes where recombination is slower, often leading to a positive ionospheric storm (TEC increase). In contrast, the storm-induced circulation from high to low latitudes transports molecular-rich air ($[O_2]$, $[N_2]$), increasing the $\frac{[N_2]}{[O]}$ ratio. This change in atmospheric composition enhances the loss rate of atomic oxygen ions (O^+) through recombination, which can lead to a negative ionospheric storm (TEC decrease) [Vijaya Lekshmi et al., 2011, Prölss, 1995].

Both PPEF and DDEF were studied by Fejer et al. [1983], who observed their effects using incoherent scatter radar at Jicamarca latitude. Later, Amory-Mazaudier [1982] detected DDEF above Saint-Santin latitude, with results consistent with the disturbed dynamo model. On the other hand, Le Huy and Amory-Mazaudier [2005] modeled by the first time the magnetic field associated with ionospheric disturbances, assuming that, for mid-low latitudes, they are mainly driven by DDEF and PPEF. They considered that during the MP, PPEF magnetic effects are dominant, whereas during the recovery phase (RP), regional magnetic field variations are mainly driven by DDEF. Following this initial approach, several studies [Le Huy and Amory-Mazaudier, 2008, Zaka et al., 2009] have focused on analyzing local responses at different latitudes and longitudes to analyze the different local responses associated to DDEF phenomena. In Liu et al. [2013], a statistical study was conducted to understand ionospheric responses as a function of local sectors during GS onset. Later, Younas et al. [2020] proposed using Fourier analysis to isolate magnetic fluctuations related to DDEF and PPEF based on their quasi-periodic behavior, identifying periods of approximately 24 hours for DDEF and less than 4 hours for PPEF.

In case of Mexico, regional space weather has a relatively recent history. On 2014, it was formed the Mexican Space Weather Service (SCIESMEX), leading to combined efforts to understand the regional space weather phenomena, in the middle of a global perspective Gonzalez-Esparza et al. [2017]. From this teamwork, studies like Corona-Romero et al. [2018, 2017] have been focused on local geomagnetic response during GS periods, and the computing of a local K index specific for Mexico. On the other hand, Romero-Hernandez et al. [2017], Sergeeva et al. [2017], Sergeeva et al. [2020] found important ionospheric activity related to geomagnetic storms. In Castellanos-Velazco et al. [2024], the main focus was analyzing the local geomagnetic response related to PPEF and DDEF phenomena, during 20 strong GS, from 2003 to 2018, where one of the main conclusions were the need of comparing the geomagnetic response in the center of Mexico with different locations at similar geomagnetic latitudes.

In this paper, we analyze the PPEF magnetic signatures at different longitude sectors, looking for the sector where these fluctuations are more intense. We compare the results with the Asymmetric Ring Current activity in order to analyze its influence on PPEF strength and so, on local geomagnetic response. We selected the GS of March 17th, 2015 which is also known as the St. Patrick's event. This GS is known to be the most intense GS during the solar cycle 24th, reaching -274 nT during its MP.

2 St. Patrick Geomagnetic Storm

The geomagnetic storm (GS) analyzed in this work is the intense event that occurred on March 17, 2015, widely known as the St. Patrick's Day storm. It is considered the most intense storm of solar cycle 24 Wu et al. [2016]. The storm was triggered by a coronal mass ejection (CME) associated with a C9.1 solar flare and a subsequent dimming observed on March 15. This CME, recorded as a partial halo by the SOHO/LASCO coronagraphs, had an estimated speed of approximately 660 km/s.

Figure 1 provides an overview of the interplanetary conditions that led to the St. Patrick's Day storm. A more detailed description about the trigger event is provided by Wu et al. [2016]. Panel (a) shows the total interplanetary magnetic field, B_T (black line), and its north-south component, B_Z (red line). Panels (b) and (c) presents solar wind properties, including the induced Interplanetary Electric Field (E_Y), calculated as $-V_X \times B_Z$; and dynamic pressure. Finally, panel (e) displays the geomagnetic response through the SYMH and ASYH indices (green and yellow lines respectively).

The key milestones of the event are marked by three vertical dashed lines in Figure 1. The first line (left) indicates the arrival of an interplanetary shock (IPS) at 04:30 UT at the L1 point. The IPS signature is clear from the step-like increases in B_T , E_Y and P_{dyn} . This shock compressed the magnetosphere, causing a Sudden Storm Commencement (SSC) visible as a sharp increase in the SYMH index between 04:30 and 07:00 UT, highlighted by light gray shading.

The region following the shock, the CME sheath, is identified by highly fluctuating B_T and B_Z , enhanced P_{dyn} . A period of southward B_Z between 06:00 and 09:00 UT initiated the what we called, the first stage of main phase (MP), driving the SYMH index down to its first minimum of -80 nT.

The second dashed line (14:00 UT) marks the arrival of the magnetic cloud, identified by Wu et al. [2016] as the main driver of the GS. Its passage is identified by a smooth, strong rotation of the magnetic field, a drop in temperature, and so, a corresponding low plasma beta. A sustained and intense southward B_Z component (reaching -40 nT) within the cloud marked the beginning of the second (and more intense) stage of the MP. The SYMH index decreased sharply, reaching its global minimum of -234 nT at 22:47 UT. The entire main phase period is shaded in dark gray in panel (e).

The third dashed line (23:00 UT) indicates the trailing edge of the magnetic cloud and the entry into a high-speed solar wind (HSSW) stream. While the high solar wind speed persists, the geomagnetic activity subsides as B_Z turns northward and the E_Y drops close to zero, ending the geoeffective driving of the storm.

Since we are interested in the effects of the PRC during its highest activity altogether with PPEF effects, we will focusing on a time window of a few hours during MP. This time window is going to be in function of periods when $E_Y > 0.5 mVm^{-1}$ and $P_{dyn} > 29.71 nPa$. The first criteria is based on the discussion in Burton et al. [1975], where they find the energy injection during MP is dependent of E_Y , setting $E_{Y0} = 0.5 mVm^{-1}$ as a minimum value. On the other hand, Xie et al. [2008] set $P_{dyn} > 15 nPa$ as a good estimator for a geoeffective dynamic pressure for strong geomagnetic storms. Considering that 15 nPa is a very low threshold for the period between march 15 - 17 as it can be seen in the Figure, we set the threshold through computing the 95% of Cumulative Density Function (CDF). Thus, we obtained a threshold of 29.71 nPa. Hence, obtain two possible time windows: 6-7 UT and 13-18 UT. Since the second period corresponds to the highest ASYH values, we selected this period for study, and its highlighted by the violet shading. Hence, during the selected time window, we expect higher injection and Electric Field penetration (under shielding) and PRC activity (Over shielding), which implies higher related magnetic fluctuations.

3 Procedure

3.1 Geomagnetic Data

The geomagnetic data analyzed in this paper were downloaded from the INTERMAGNET public database (*International Real-Time Magnetic Observatory Network* [INTERMAGNET, 2021]), corresponding

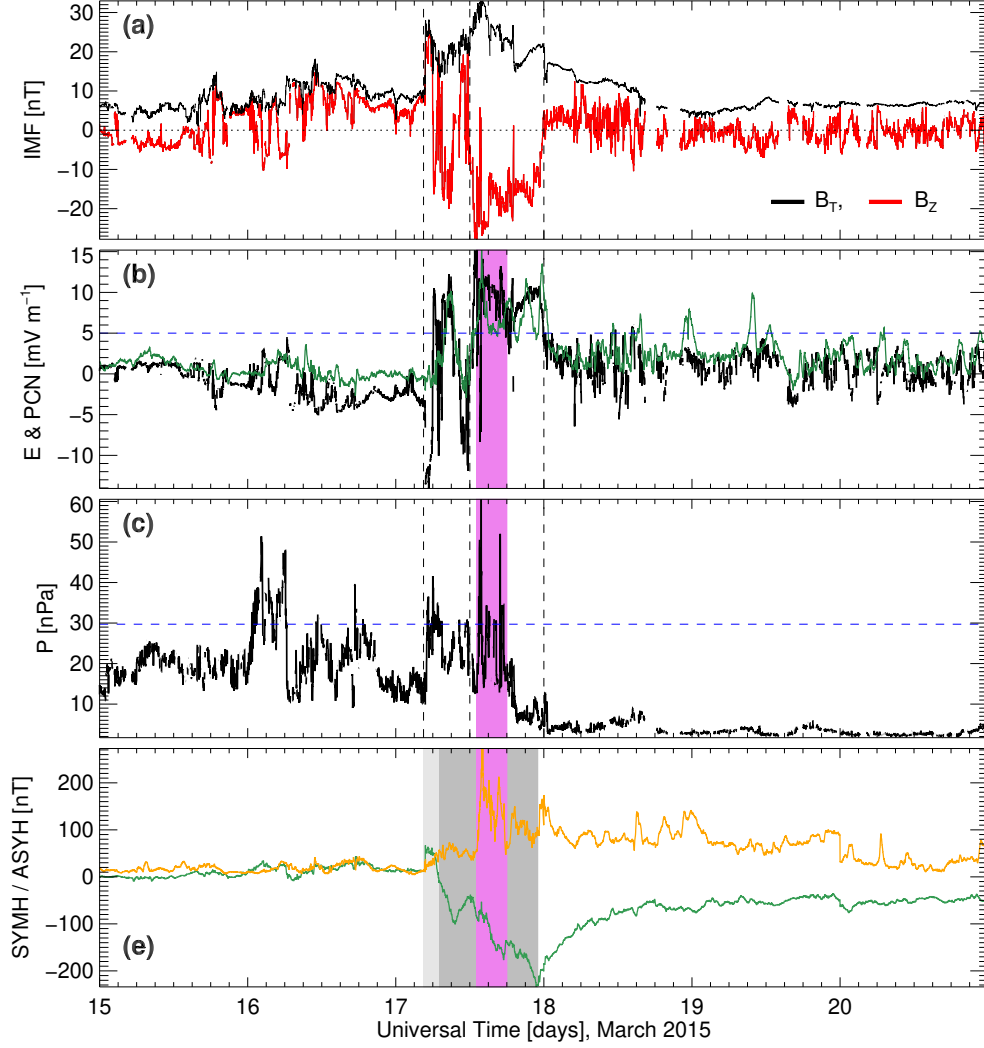


Figure 1: Interplanetary measurements. Panel (a): interplanetary magnetic field total component (black line) and z component (red line). Panel (b): E_Y Interplanetary Electric Field (blue line). Panel (c): Dynamic Pressure [nPa]. Panel (d): SYMH (green line) and ASYH index (orange line), highlighting the initial phase (light gray shade), main phase 1 (medium gray shade) and main phase 2 (dark gray shade).

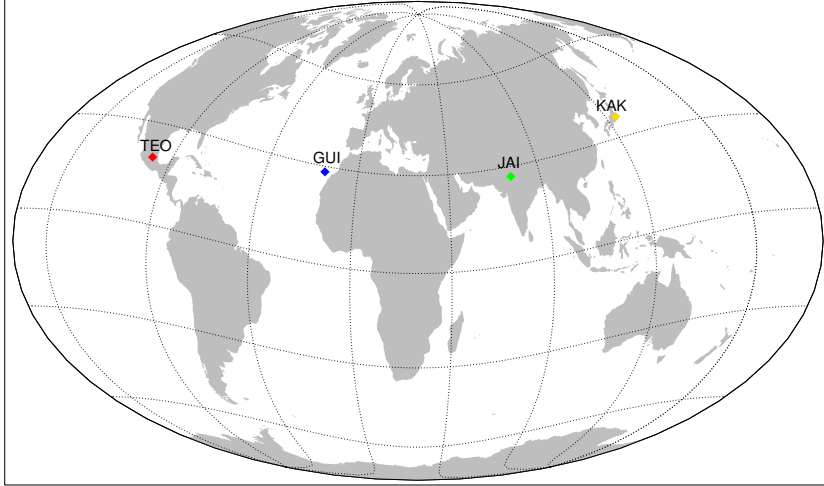


Figure 2: Geographic position of the four magnetic OB, distributed at $70 - 90^\circ$ from each other

to the observatories (OB) listed in Table 1.

Table 1: List of observatories considered for this study. Each OB is separated longitudinally at $70 - 90^\circ$ apart. The OB selected for the main analysis are light gray shaded.

Observatory	IAGA code	Country/ Territory	Latitude		Longitude	
			Geographic	Magnetic	Geographic	Magnetic
Beijing	BMT	China	40.3 N	30.53 N	116.2 E	172 W
Cheongyang	CYG	South Korea	36.37 N	27 N	126.854 E	162.43 W
Guimar	GUI	Spain	28.317 N	33.21 N	16.433 W	61.1 E
Honolulu	HON	USA	21.317 N	21.49 N	158 W	90.03 W
Jaipur	JAI	India	26.917 N	18.42 N	75.8 E	150.47 E
Kakioka	KAK	Japan	36.232 N	27.8 N	140.186 E	149.73 W
Lanzhou	LZH	China	36.083 N	26.23 N	103.833 E	176.79 E
San Juan	SJG	USA, Puerto Rico	18.11 N	28.19 N	66.15 W	6.1 E
Tamanraset	TAM	Argelia	22.783 N	24.3 N	5.571 E	82.29 E
Teoloyucan	TEO	Mexico	19.747 N	27.81 N	99.182 W	28.4 W

To study the different geomagnetic responses, we focused on data from observatories (OB) listed in Table 1. As it will be seen later in Section B, OB which are longitudinally too close, will have similar results. Thus, we focus in the four OB highlighted in the Table, which are longitudinally separated by $60 - 90^\circ$ from each other and are highlighted in Figure 1. As mentioned before, the analysis will be focused during a period of maximum injection, defined by a large E_Y interplanetary electric field penetration and high dynamic pressure and maximum activity from the partial ring current (PRC), measured by ASYH index (273 nT). Both mechanisms imply the under shielding and overshielding of the $PPEF$, thus we could expect high H_{PPEF} fluctuations during this period, from 14:07 to 19:00 UT (5 hours) and shortly after, from 19:00 UT to 23:00 UT which coincides with a period of high magnitude E_Y but low P_{dyn} (*approx 6 nPa*). Interestingly, the period TW also coincides with each one of the selected OB being transitioning through the four local time sectors:

- Dawn-Noon sector (TEO), 06:01 - 12:00 LT.
- Noon-Dusk sector (GUI), 12:01 - 18:00 LT.
- Dusk-Midnight sector (JAI), 18:01 - 00:00 LT.
- Midnight-Dawn sector (KAK), 00:01 - 06:00 LT.

3.2 Magnetic fluctuations of ionospheric origin

It is well accepted that [Rishbeth and Garriott, 1969, Knecht and B.M, 1985, Gjerloev, 2012, van de Kamp, 2013] local recordings H_{loc} of a ground magnetometer can be considered a linear combination of different geomagnetic sources:

$$H_{loc} = H_{SQ} + H_0 + H_{MT} + H_{others} + H_I, \quad (1)$$

where H_{SQ} represents the diurnal variation and H_0 is the main field during a month window. H_{MT} represents the variations induced by the intensification of magnetospheric currents during the GS. In the case of H_{others} , it corresponds to magnetic sources less significant, like (and no limited to)) magnetotellurics, crustal magnetization, etc; which can be neglected for not being considered as relevant compared to the others mentioned. Finally, H_I Le Huy and Amory-Mazaudier [2005] corresponds to the magnetic variations driven by ionospheric current activity during the GS. Thus, the local geomagnetic response during a GS is given by:

$$\Delta H = H_{loc} - (H_{SQ} + H_0). \quad (2)$$

We provide a detailed explanation of how we model the diurnal variation and the monthly baseline in Appendix A.

From Le Huy and Amory-Mazaudier [2005], Amory-Mazaudier et al. [2017], Younas et al. [2020], for mid-low magnetic latitudes, with λ as the geomagnetic latitude of each OB, we can roughly approximate H_{MT} as $H_{MT} \approx SYMH \cdot \cos(\lambda)$. Thus, H_I can be approximated as:

$$H_I = H_{DDEF} + H_{PPEF} + H_{others} \approx H - (H_{SQ} + H_0 + SYM - H \cdot \cos(\lambda)), \quad (3)$$

where H_{DDEF} and H_{PPEF} represent the magnetic fields associated with the disturbed dynamo electric fields and the prompt penetration electric fields, respectively. H_{others} on the other hand remains as a noise.

In order to isolate H_{DDEF} and H_{PPEF} from each other, we proceed under the assumption that both ionospheric currents drive quasi-periodic fluctuations [Nishida, 1968a, Blanc and Richmond, 1980, Younas et al., 2020]. Hence, through a Fourier analysis, we apply pass-band filters for frequencies between $69.5 - 550\mu\text{Hz}$ (corresponding to periods in 4 hours to 30 minutes range).

Given the role of PRC within local geomagnetic response, Younas et al. [2020] applies an extra step, where:

$$H'_I = H_I - ASYH \cdot \cos(\lambda), \quad (4)$$

in order to remove also the contribution from ASYH index. This suggest that PRC effects remains within H_I , specially in the night side and early morning [Takahashi et al., 1991]. However, given the short duration of PRC activity, in comparison with the symmetric part of the system, combined with the limited longitudinal extension of the PRC would mean that Eq 4 shouldn't be applied equally for the OB, specially when this OB are located by $60 - 90^\circ$ degrees from each other.

Hence, the analysis will be mainly focused on comparing the result of H_I with ASYH index. Given the order of magnitude of ASYH, compared with H_{PPEF} (and H_{DDEF}), this would mean that ASYH will be the main source for H_I , depending on the OB position with respect to local Time.

4 Results

In the following section, we focus on the results related to the geomagnetic response to the St. Patrick's GS for each OB listed in Table 1. Then we show the results of Equation 3 for isolating H_I , and an analysis related to local time sector in which the GS developed as well as a comparison between each OB result.

In Figure 3, we show in top panel, the ASYH index, and the computed ionospheric magnetic contribution H_I (red lines) for GUI, JAI, KAK and KAK (second to fifth panels). The dashed vertical

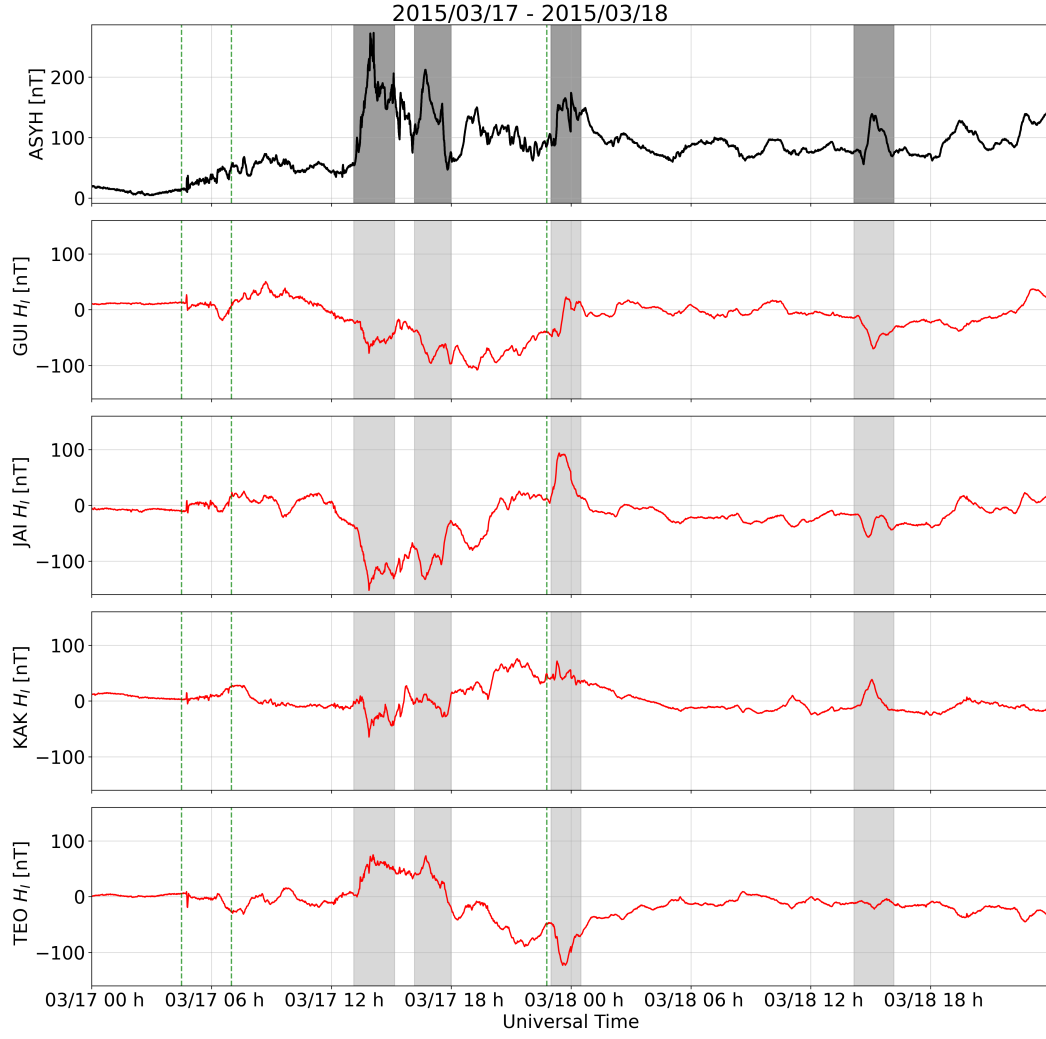


Figure 3: Comparison between ASYH index (top panel) with H_I from Eq 3, downward: GUI, JAI, KAK and TEO. The green dashed vertical lines represent the beginning of SSC, MP and RP.

lines represent the transition of quietness-SSC, SSC-MP and MP-RP. Our main attention will be on the MP and more specifically, during and after the period of maximum injection determined by TW.

As mentioned at the end of the last section, we will focus on periods where ASYH could be the main source for each H_I . The easiest way to observe this is acknowledge periods when H_I presents a similar/opposed pattern to ASYH. This is mostly seen during and after TW. Notice how, between 13:00 and 16 UT the behavior of H_I seems to be opposed to ASYH from GUI to KAK, specially in JAI. In TEO on the other hand, the pattern seems to be more alike to ASYH. However, as time pass, this oppose/similar pattern is lost in most places, until around UT, where the situation is inverse. Now, TEO presents a response apparently opposed meanwhile the other 3 OB presents a response more similar to ASYH

We begin by analyzing the magnetic data from GUI. During the TW, GUI was on the noon-dusk sector (11:00-17:00 LT). The peak ASYH enhancement (274 nT) occurs at 12:07 LT at GUI, nearly coinciding with the noon sector. ΔH shows a smooth decrease during TW, while H_I appears to exhibit a pattern opposite to ASYH. In the period right after TW, ΔH maintains sustained minimum values around -212 nT until 21:00 LT (23:00 UT), when it suddenly increases concurrently, coinciding with an ASYH enhancement. TEC response on the other hand presents an enhancement period around 16:00 LT (18:00 UT), of $\sim 55\%$ TECu, followed by a second enhancement of $\sim 58\%$ almost at 0 LT.

In case of JAI OB, TW occur during night hours. The peak ASYH enhancement occurs at 19:07 LT, while both ΔH and H_I show a more pronounced decrease compared to GUI. ΔH reaches its minimum nearly simultaneously with the second ASYH pulse. During TW, H_I seems to behave opposed to ASYH. TEC response on the other hand, is does not present a significant differences until march 18. During this day, we observe a consistent negative response of around 70%.

At KAK OB, ΔH shows a significantly less pronounced decrease during TW compared to the previous observatories. The minimum ΔH is reached at approximately 02:40 LT. H_I also shows a pattern opposite to the ASYH index, though this is only evident until 01:00 LT. After this time, H_I gradually appears to follow a pattern more similar to ASYH. After TW, H_I maintains intense negative values between -90 nT and -150 nT. There is no apparent response in ΔH during the final ASYH pulse beginning at 08:00 LT. Conversely, H_I shows a growth phase during and after TW, reaching a maximum value shortly after dawn at 06:00 LT. In ΔTEC , similar to JAI's case, an enhancement which is consistent with the first stage of MP. Nevertheless, in this case this enhancement is presented during night hours, conforming bulb, or like a second peak (seen in Figure 6). Next, there is a negative response of 50% the next day of MP.

For TEO, which represents the opposite case of JAI, we observe that TW occur during day-side hours. ΔH presents an almost linear descending trend during and after TW until it reaches its minimum at approximately 16:20 LT, consistent with the timing of an ASYH pulse. H_I , on the other hand, shows a pattern more similar to ASYH behavior during TW, with positive values that contrast with previous cases and peak values occurring approximately simultaneously with ASYH pulses. After TW, however, H_I begins a decreasing phase, reaching a minimum at the same time as ΔH and the final ASYH enhancement. Examining the TEC relative response, we observe a sharp enhancement of 70% near the end of TW at 11:00 LT.

5 Discussion

From Figure 3, we observe significant variations in the results depending on the local time sector of each observatory during TW. More specifically, we acknowledge variations in the partial ring current (PRC) influence on the local response. When comparing the geomagnetic responses, we observe that H_I behave more similar or opposed to ASYH which means that PRC contributed more to ΔH . These results might be one of the reasons about ΔH computed in GUI, JAI, KAK and TEO, have different decrease behaviors during TW, particularly around 14:07 UT (corresponding to 12:07, 19:07, 23:07 and 07:07 LT respectively).

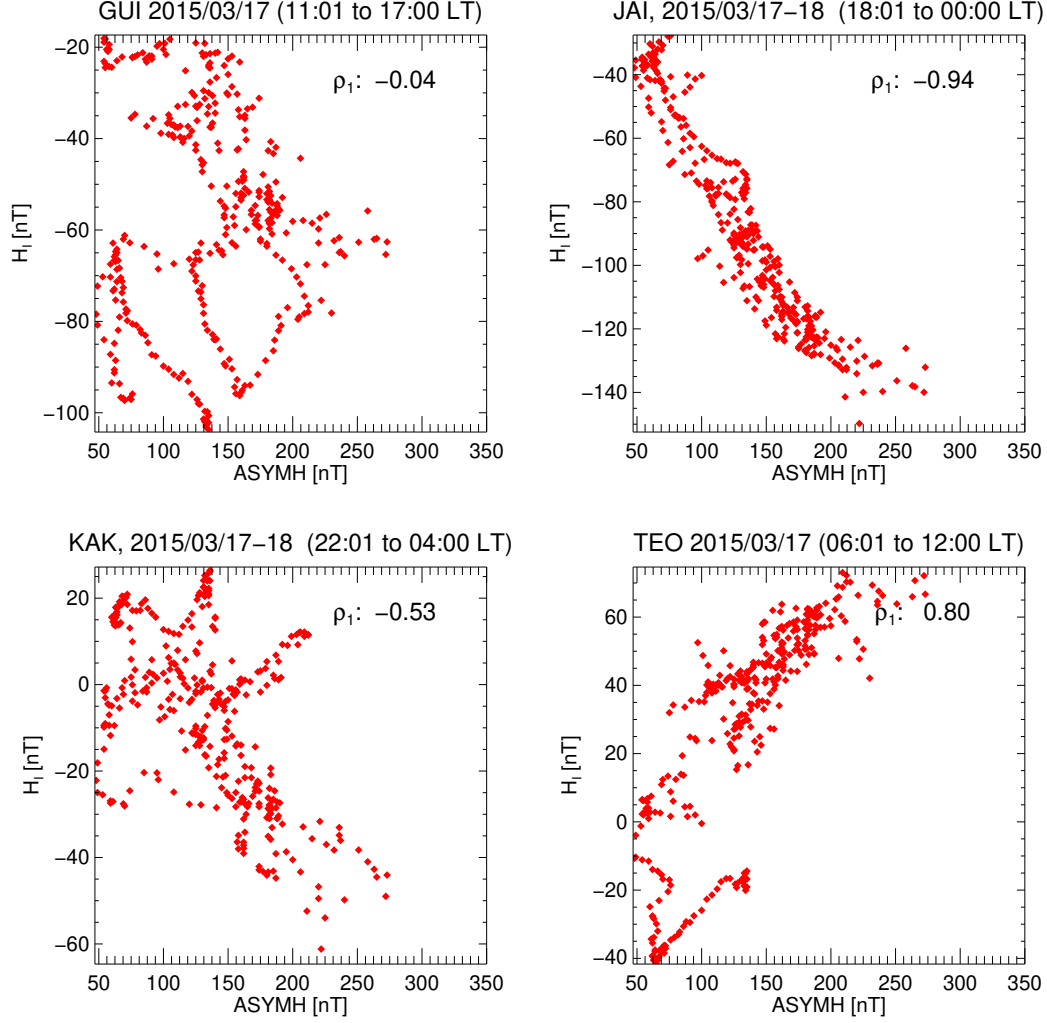


Figure 4: Correlation Analysis ASYH and H_I , during the period 1300 - 1900 UT, which is consistent with the period of highest ASYH activity

From Kalegaev et al. [2008], we consider that PRC presents a significant but also short contribution to the local geomagnetic response. Given the Eq 3, its clear that if PRC affects ΔH , will surely affect the resulting H_I , as Younas et al. [2020] implied by adding Eq 4, in order to remove the effects of PRC. Thus, we can take all those periods when H_I follows a similar/opposed pattern of ASYH, then Eq 3 should be considering adding a term for ASYH, as done in Younas et al. [2020]. However, as mentioned before, this is not the case for all the OB.

To quantitatively confirm weather PRC contribution remains in ΔH and so, presents a noise contribution factor within H_I , We analyze the correlation between H_I and ASYH, focusing on TW (13:00-19:00 UT), during which PRC activity occurs. Later, given the relation PRC - FAC - PPEF, we analyze the H_{PPEF} fluctuations distribution during MP, looking for the sectors in order to observe weather or not PRC activity can contribute to the PPEF intensity.

First, we computed the Pearson correlation coefficient (ρ) in two stages: first, ρ_1 for ASYH vs H_I during TW, and second, ρ_2 for ASYH vs H_I before the GS. The results are shown in Figure 4, where red diamonds represent ρ_1 (before removing H_{PPEF}), and gray diamonds represent ρ_2 .

According to Makarov [2022], it is implied that there is a strong anti-correlation between ASYH and SYMH indices. Therefore, a strongly positive or negative ρ coefficient can be interpreted as ΔH being influenced by PRC activity, so $H_I = H_{DDEF} + H_{PPEF} + PRC$. In this context, a strongly positive ρ implies a negative contribution to the local H component; in other words, the PRC may intensify the

local geomagnetic response. Conversely, a strongly negative ρ suggests that the PRC may attenuate the local geomagnetic response.

The two observatories showing strong correlation (or anti-correlation) are JAI and TEO. Interestingly, these were also the observatories closest to the dusk and dawn sectors, respectively, at the beginning of TW.

Following with the influence of PRC on H_{PPEF} fluctuation amplitudes, we carried out an analysis of H_{PPEF} distribution magnitudes, from march 17th to march 18th, shown in Figure 5. From top panels, we observe that H_{PPEF} magnitudes (blue bars) in each OB seems to approach to a Gaussian distribution (fitted with a red line). All OB present media $\mu \sim 0$ nT and a very small skewness, seen at first glance, being JAI the farther from 0 ($\mu = 0.67$). The standard deviation σ seen in each OB has a maximum difference goes from 5.2 to 8.15, being not so significant.

Both, the almost symmetry of the distribution and the center near zero are consistent with H_{PPEF} having a periodic behavior. The shape of the distribution on the other hand, so as the length of both tails are a good indicator of the intensity of these fluctuations. From here, it can be seen that the longest tails corresponds to JAI and TEO which is consistent with JAI and TEO having higher peaks of H_{PPEF} amplitudes. GUI and KAK on the other hand, present greatest and minor dispersion respectively. In case of GUI, this is consistent with H_{PPEF} amplitudes not having a noticeable contrast during TW, meanwhile KAK presents minor activity in general, but with a higher contrast of amplitudes during TW.

From here, we are interested in estimating the point at which H_{PPEF} can be considered as extreme, for each OB. We refer this point as *Threshold*. These thresholds can be used for determining periods of more significant PPEF activity. Thus, the higher the threshold, higher PPEF activity in a certain region.

First, from the good symmetry in all distributions, we can consider the use of an absolute value threshold. From this approach, we set each threshold as the 90% of each cumulative H_{PPEF} probability distribution function, shown in bottom panels of Figure 5 (blue lines). The threshold is indicated by the red dashed line for the 2 days period.

The highest thresholds correspond to JAI (28.63 and 40.98 nT) followed by TEO (20.31 and 33.79), meanwhile, in GUI and KAK, have significantly minor thresholds. What this difference may indicate is how H_{PPEF} intensity is affected by the LT sector at the moment of the highest ASYH activity. This result is consistent with the results of $\rho_{1,2}$ for these OB, which might reinforce the idea of H_{PPEF} being influenced by PRC activity, indicating not just the impact of PRC activity but also on the proximity of JAI and TEO to the dusk-midnight sectors during the highest PRC activity.

Regarding the ionospheric response, we observe that the TEO and GUI regions (monitored by the UCOE and LPAL receivers) present a positive storm effect, while the JAI and KAK regions (monitored by the NPGJ and TSK2 receivers) present a negative effect. From Thomas and Venables [1966], Vijaya Lekshmi et al. [2011], it is understood that positive phases at mid-low latitudes are often initiated on the day-side during the main phase of the storm. The physical mechanism for this is complex and involves thermospheric winds, $\mathbf{E} \times \mathbf{B}$ drift, and, most importantly for this study, the presence of eastward PPEF. According to Vijaya Lekshmi et al. [2011], the influence of PPEF alone may not always induce large TEC enhancements; its effects are often combined with neutral wind circulation and $\mathbf{E} \times \mathbf{B}$ drift to produce significant positive ionospheric storms.

The negative effects, on the other hand, are more related to regions that were on the night-side during the storm's main phase, which is the case for JAI and KAK in our event. These negative phases, however, typically manifest the following morning in local time. Studies like Fuller-Rowell et al. [1994], Prölss [1995] describe the mechanism responsible for these negative effects. It involves a storm-induced circulation that pushes molecular-rich air ($[\text{N}_2]$, $[\text{O}_2]$) from high latitudes toward lower latitudes. This increases the $\frac{[\text{N}_2]}{[\text{O}]}$ ratio, which in turn enhances the loss rate of atomic oxygen ions (O^+) through recombination. Although this circulation begins during the night, its effects peak the next morning when photo-ionization commences but is overwhelmed by the heightened recombination rate, resulting in an electron density depletion and a negative TEC storm.

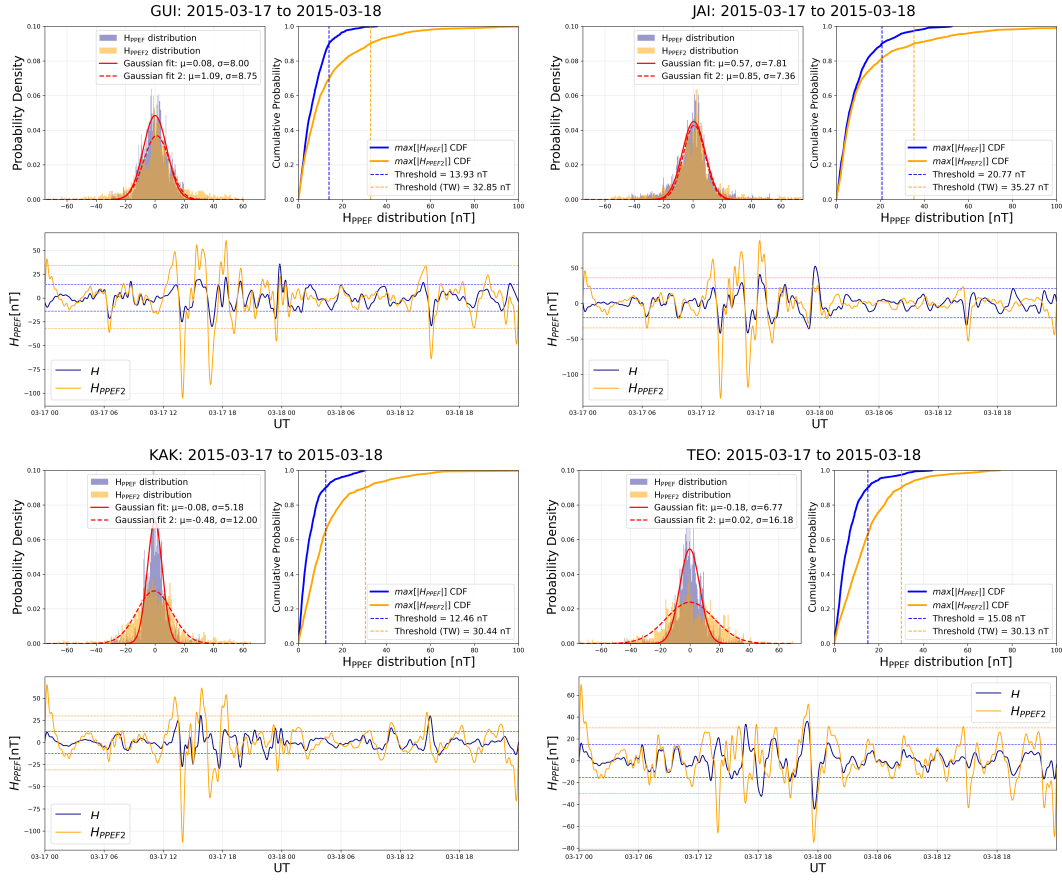


Figure 5: H_{PPEF} distribution for each OB. Color dashed vertical lines represent the a threshold for H_{PPEF} to be considered as intense amplitudes

6 conclusions

Within this project, we analyzed longitudinal magnetic fluctuations driven by Prompt Penetration Electric Fields (PPEF) during the intense St. Patrick’s Day geomagnetic storm on March 17, 2015. To achieve this, we utilized geomagnetic data from ground-based observatories separated by 60–90° in longitude.

The study focused on isolating the magnetic component associated with the ionospheric current system (H_I) and the fluctuations specifically linked to PPEF (H_{PPEF}). The H_{PPEF} signal was reconstructed via pass-band filtering to retain fluctuations with periods within a range of 4h and 30 minutes. Our analysis targeted the period of enhanced activity in the ASYH index, which reflects the asymmetric part of the magnetospheric ring current system and is linked to the Partial Ring Current (PRC).

We investigated the influence of the PRC on both H_I and H_{PPEF} . These observations were complemented by analyzing relative Vertical TEC variations, considering deviations above 30% from the 27-day median as significant ionospheric perturbations. We observed positive ionospheric storm effects at the TEO and GUI regions (55% and 70% TEC increase, respectively), while the JAI and KAK regions exhibited negative storm effects (70% and 50% TEC decrease). This pattern is consistent with the local time of each region during the storm’s main phase onset, but also reflects the influence of PPEF and neutral winds flow on electron density over those regions.

By comparing H_I and H_{PPEF} with ASYH activity across different local time sectors and integrating TEC observations, we reached the following conclusions:

- PRC activity is highly correlated with fluctuations in both H_I and H_{PPEF} during the most geoeffective period of the geomagnetic storm.
- The local geomagnetic response (ΔH) is either enhanced or attenuated depending on the observatory’s proximity to the dawn or dusk sectors, where the PRC influence is strongest.
- High PRC activity significantly affects the magnitude of H_{PPEF} , particularly when an observatory is magnetically conjugate to the regions where the PRC connects to the ionosphere via Field-Aligned Currents (FACs).
- Conversely, when an observatory is located far from these PRC-FAC coupling regions (e.g., near the noon sector), the contribution to H_{PPEF} is less significant.
- The TEC enhancements at GUI and TEO are primarily driven by eastward PPEF combined with $\mathbf{E} \times \mathbf{B}$ drift from neutral wind circulation.
- The TEC depletions at JAI and KAK are attributed to storm-induced changes in thermospheric composition and circulation. The distinct mechanisms observed between the dayside and nightside regions suggest the presence of a disturbed dynamo.

7 Future work

This study constitutes the first part of a broader investigation into the local time dependence of the regional geomagnetic response. A subsequent study will focus on the effects of the Disturbance Dynamo. Furthermore, given the recent occurrence of intense geomagnetic storms, we plan to extend this analysis to events such as the ”Mother’s Day” storm and the event of October 10th. Expanding the dataset in this way will also help mitigate data quality issues, such as the limited availability of high-quality GNSS receiver data encountered in the present work.

A Processing magnetic data

It is crucial to eliminate regular variations from magnetic records to isolate magnetic contributions of interest. The main field H_0 can be removed by modeling a monthly baseline. This baseline was approximated by computing an array of daily mode values for March 2015, following Gjerloev [2012].

These daily mode values were obtained from data between 00:00 and 03:00 LT ($H_{0:3}$), then H_0 was computed by a linear least squares fitting from these daily values. To exclude data from disturbed days for the fitting, we adapted the procedure outlined in van de Kamp [2013]. Specifically, we applied a moving hourly window to calculate the hourly interquartile range (IQR). Disturbed days were identified by setting a variation threshold where:

$$IQR_{med}^i > threshold = disturbed \quad (5)$$

Here, IQR_{med}^i is the median of each daily IQR values, so if more than 50% of IQR values for a given day, surpass a determined threshold, then that day is considered as disturbed day and depreciated for the LS fitting.

The threshold was set, based on the criteria seen in Solari et al. [2017]. Basically, we analyze the monthly distribution of maximum IQR values (IQR_i^{max}) within tri-hours IQR bins, by shifting another moving window and sorting those peaks, so $IQR_1^{max} < IQR_2^{max} < \dots < IQR_n^{max}$, in order to get a cumulative IQR^{max} distribution.

Hence, we have to compute the parameters of the General Pareto Distribution (GPD) Pickands [1975] which is a type of distribution used for modeling other distributions extreme values, and it is expressed as:

$$f(x) = \begin{cases} \frac{1}{\sigma} \left(1 + k \frac{x-u}{\sigma}\right)^{-1-\frac{1}{k}} & \text{if } k \neq 0 \\ \frac{1}{\sigma} \exp\left(-\frac{x-u}{\sigma}\right) & \text{if } k = 0 \end{cases} \quad (6)$$

where the three GPD parameters are shape k , scale σ and location u for the cumulative IQR_i^{max} . The last parameter, u is the one which can be considered as a threshold point [Hosking, 1990, Solari et al., 2017]. Then, for each IQR_i^{max} , we compute $[k_i, \sigma_i, u_i]$, then we apply a modified Anderson-Darling test. The threshold, u_t , is considered once the test gives the most optimum validation result.

To compute the diurnal variation, we identified what we refer to as local quiet days (LQD) for each station. Following the methodology in van de Kamp [2013], we considered the five days with the smallest daily IQR peaks as LQDs. We then computed the average of the H component data during these five quietest days:

$$\langle LDQ_H(t) \rangle = \frac{1}{5} \sum_{i=1}^5 H_i(t) - \langle H_i 0 : 3 \rangle \quad (7)$$

where we normalize the offset of each daily data by subtracting the average of the first four hours of magnetic data in local time, as indicated by the second term in Equation 7. Subsequently, we model the diurnal variation H_{SQ} by applying a low-pass filter to the Fourier Transform of the result from Equation 7, following the procedure in Equation (1) of van de Kamp [2013].

B ionospheric and magnetic response

B.1 Total Electron Content

The overall ionosphere dynamics in each considered region were characterized by the vertical Total Electron Content (TEC) variations in the vicinity of the chosen magnetic OB. TEC values were derived from GNSS measurements using TayAbsTEC method Yasyukevich et al. [2015a,b], Mylnikova et al. [2015]. In particular, data from UCOE, LPAL, NPGJ and TSK2 receivers was used to obtain TEC values for TEO, GUI, JAI and KAK regions, respectively.

In Figure 6, it is shown the TEC responses (red line) during the period of the GS, compared with the median of the 27 days TEC prior to the GS (blue line). However, given the different location of the GNSS stations, it is usual to derive different expected daily ranges (in GUI/LPAL is around 16-70 TECu; JAI/NPGJ is 20 - 80 TECu; KAK/TSK2 is 7 - 38; TEO/UCOE is 10 - 55 TECu). The same goes for different amplitude responses over/bellow the median, so checking only the TECu magnitude may not be possible to compare the results between the different GNSS stations.

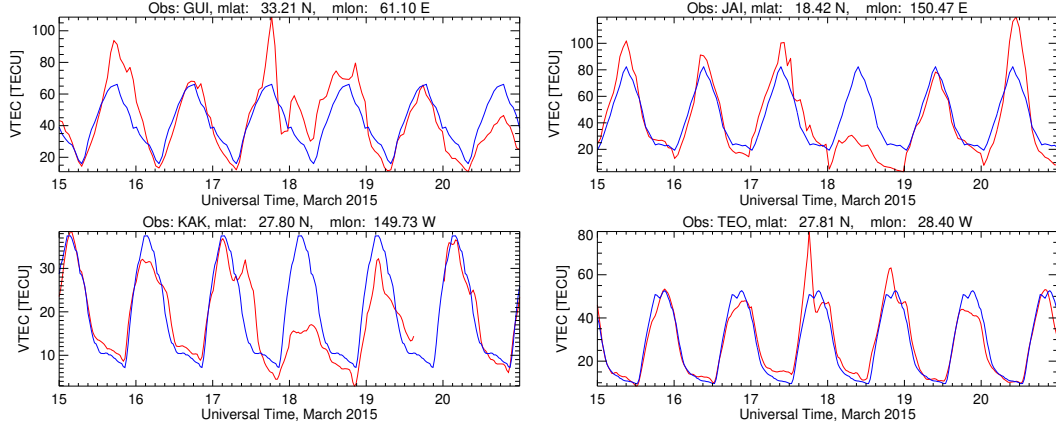


Figure 6: Total Electron Content from LPAL (GUI), NPGJ (JAI), TSK2 (KAK) and UCOE (TEO)

Another issue is that we have to consider that the open data obtained for this work, doesn't have the best quality, even more considering the period of time. The quality of data is a sum of errors which depend on the satellite swarm, so as the receptor itself. We could smooth some of these errors but couldn't be avoided at all. Considering also that we are studying results from points in different longitudinal sectors (Mexico, Canarias Islands, India and Japan), there can be issues by comparing absolute TEC values. To deal with this issue, we instead analyze relative values, defined by:

$$\Delta TEC = \frac{TEC_{obs} - TEC_e}{TEC_e}, \quad (8)$$

where TEC_{obs} is the observed response during the period of interest and TEC_e is the expected TEC, derived from the 27 days median TEC. ΔTEC is then, measuring TEC percentage difference between the expected value and the TEC observed during the GS Vijaya Lekshmi et al. [2011], Sergeeva et al. [2017].

Acknowledgements The authors gratefully acknowledge the data and services provided by various institutions that made this work possible. Geomagnetic indices (ASY/SYM-H) were obtained from the International Service of Geomagnetic Indices (ISGI). Global geomagnetic data from the GUI, JAI, and KAK observatories were provided by INTERMAGNET, while local data from the TEO Observatory were sourced from LANCE-SciesMEX, REGMEX, and the Magnetic Service. GNSS data for the LPAL and TSK2 stations were accessed via the International GNSS Service (IGS), and data for the NPGJ station were provided by the NSF SAGE facility operated by EarthScope Consortium (under NSF award 1724509).

We extend our special thanks to the personnel of the Servicio de Geodesia Satelital, particularly Luis Salazar Tlaczani, and the TLALOCNet team for the maintenance of the UCOE station, data acquisition, and IT support. The GNSS data from this network, essential for TEC calculations, are supported by project PAPIIT IN111225 (E. Cabral-Cano).

Finally, this research was funded by the Universidad Nacional Autónoma de México (UNAM) and the Secretaría de Educación, Ciencia, Tecnología e Innovación (SECIHTI).

References

- R. J. Strangeway C.T Russell, J. G. Luhman. *SPACE PHYSICS AN INTRODUCTION*. Cambridge university press. University Pringting House, Cambridge CB2 (BS, United Kingdom, 2016. ISBN 978-1-107-09882-4.
- Atsuhiro Nishida. Geomagnetic D_p 2 fluctuations and associated magnetospheric phenomena. *Journal of Geophysical Research*, 73(5):1795–1803, March 1968a. doi: 10.1029/JA073i005p01795.
- Atsuhiro Nishida. Coherence of geomagnetic DP 2 fluctuations with interplanetary magnetic variations. *Journal of Geophysical Research*, 73(17):5549–5559, September 1968b. doi: 10.1029/JA073i017p05549.

- M. Blanc and A. D. Richmond. The ionospheric disturbance dynamo. *Journal of Geophysical Research*, 85(A4):1669–1686, April 1980. doi: 10.1029/JA085iA04p01669.
- Vytenis M. Vasyliunas. Mathematical models of magnetospheric convection and its coupling to the ionosphere. In B. M. McCormac, editor, *Particles and Fields in the Magnetosphere*, pages 60–71, Dordrecht, 1970. Springer Netherlands. ISBN 978-94-010-3284-1.
- R. K. Jaggi and Richard A. Wolf. Self-consistent calculation of the motion of a sheet of ions in the magnetosphere. *Journal of Geophysical Research*, 78:2852–2866, 1973. URL <https://api.semanticscholar.org/CorpusID:121315484>.
- Yong Wei, Biqiang Zhao, Guozhu Li, and Weixing Wan. Electric field penetration into earth’s ionosphere: a brief review for 2000–2013. *Science Bulletin*, 60(8):748–761, 2015. ISSN 2095-9273. doi: <https://doi.org/10.1007/s11434-015-0749-4>. URL <https://www.sciencedirect.com/science/article/pii/S2095927316304376>.
- M. C. Kelley, B. G. Fejer, and C. A. Gonzales. An explanation for anomalous equatorial ionospheric electric fields associated with a northward turning of the interplanetary magnetic field. *Geophysical Research Letters*, 6(4):301–304, April 1979. doi: 10.1029/GL006i004p00301.
- D. Vijaya Lekshmi, N. Balan, S. Tulasi Ram, and J. Y. Liu. Statistics of geomagnetic storms and ionospheric storms at low and mid latitudes in two solar cycles. *Journal of Geophysical Research (Space Physics)*, 116(A11):A11328, November 2011. doi: 10.1029/2011JA017042.
- G. W. Prölss. *Ionospheric F-region storms*, volume 2, pages 195–248. CRC Press, 1995. doi: 10.1201/9780203713297.
- B. G. Fejer, M. F. Larsen, and D. T. Farley. Equatorial disturbance dynamo electric fields. *Geophysical Research Letters*, 10(7):537–540, 1983. doi: <https://doi.org/10.1029/GL010i007p00537>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/GL010i007p00537>.
- Christine Amory-Mazaudier. Electric currents above saint-santin. i - data. *Journal of Geophysical Research*, 87:2459–2464, 04 1982. doi: 10.1029/JA087iA04p02459.
- Minh Le Huy and Christine Amory-Mazaudier. Magnetic signature of the ionospheric disturbance dynamo at equatorial latitudes: “ D_{dyn} ”. *Journal of Geophysical Research (Space Physics)*, 110(A10):A10301, October 2005. doi: 10.1029/2004JA010578.
- Minh Le Huy and Christine Amory-Mazaudier. Planetary magnetic signature of the storm wind disturbance dynamo currents: D_{dyn} . *Journal of Geophysical Research (Space Physics)*, 113(A2):A02312, February 2008. doi: 10.1029/2007JA012686.
- K. Z. Zaka, A. T. Koba, P. Assamoi, O. K. Obrou, V. Doumbia, K. Boka, B. J. P. Adohi, and N. M. Mene. Latitudinal profile of the ionospheric disturbance dynamo magnetic signature: comparison with the DP2 magnetic disturbance. *Annales Geophysicae*, 27(9):3523–3536, September 2009. doi: 10.5194/angeo-27-3523-2009.
- J. Y. Liu, W. H. Yang, C. H. Lin, Y. I. Chen, and I. T. Lee. A statistical study on the characteristics of ionospheric storms in the equatorial ionization anomaly region: GPS-TEC observed over Taiwan. *Journal of Geophysical Research (Space Physics)*, 118(6):3856–3865, June 2013. doi: 10.1002/jgra.50366.
- W. Younas, C. Amory-Mazaudier, Majid Khan, and R. Fleury. Ionospheric and Magnetic Signatures of a Space Weather Event on 25-29 August 2018: CME and HSSWs. *Journal of Geophysical Research (Space Physics)*, 125(8):e27981, August 2020. doi: 10.1029/2020JA027981.
- J. A. Gonzalez-Esparza, V. De la Luz, P. Corona-Romero, J. C. Mejia-Ambriz, L. X. Gonzalez, M. A. Sergeeva, E. Romero-Hernandez, and E. Aguilar-Rodriguez. Mexican space weather service (sciesmex). *Space Weather*, 15(1):3–11, 2017. doi: <https://doi.org/10.1002/2016SW001496>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2016SW001496>.
- P. Corona-Romero, E. Sanchez-Garcia, M. Sergeeva, J.A. Gonzalez-Esparza, G. Cifuentes-Nava, E. Hernandez-Quintero, A. Caccavari, E. Aguilar-Rodriguez, J.C. Mejia-Ambriz, V. de-la-Luz, and L.X. Gonzalez. Kmex: the mexican geomagnetic k index. In *Proceedings of the II Pan-American Workshop on Geomagnetism*, pages 96–100, Basouras, Brazil, 2018.

- P. Corona-Romero, M. Sergeeva, J.A. Gonzalez-Esparza, G. Cifuentes-Nava, E. Hernandez-Quintero, and E. Caccavari, A. Sanchez-Garcia. Mexican geomagnetic k index. pages 1–6, 2017.
- E. Romero-Hernandez, J. A. Gonzalez-Esparza, M. Rodriguez-Martinez, M. A. Sergeeva, E. Aguilar-Rodriguez, J. C. Mejia-Ambriz, and V. De la Luz. Study of ionospheric disturbances over Mexico associated with transient space weather events. *Advances in Space Research*, 60(8):1838–1849, October 2017. doi: 10.1016/j.asr.2017.06.042.
- M. A. Sergeeva, O. A. Maltseva, J. A. Gonzalez-Esparza, V. De la Luz, and P. Corona-Romero. Features of TEC behaviour over the low-latitude North-American region during the period of medium solar activity. *Advances in Space Research*, 60(8):1594–1605, October 2017. doi: 10.1016/j.asr.2017.06.021.
- Maria Sergeeva, Aleksei Kalishin, Olga Maltseva, D. Blagoveshchensky, A. Gonzalez-Esparza, Pedro Corona-Romero, and Victor Gatica Acevedo. Possibility of signal reflection from the northern crest of eia: case study. pages 1–4, 03 2020. doi: 10.23919/EuCAP48036.2020.9135786.
- C.I. Castellanos-Velazco, P. Corona-Romero, J.A. González-Esparza, M.A. Sergeeva, A.L. Caccavari-Garza, and V.J. Gatica-Acevedo. Low latitude geomagnetic response associated with intense geomagnetic storms: Regional space weather in mexico. *Journal of Atmospheric and Solar-Terrestrial Physics*, 259:106237, 2024. ISSN 1364-6826. doi: <https://doi.org/10.1016/j.jastp.2024.106237>. URL <https://www.sciencedirect.com/science/article/pii/S1364682624000658>.
- Chin-Chun Wu, Kan Liou, Ronald P. Lepping, Lynn Hutting, Simon Plunkett, Russ A. Howard, and Dennis Socker. The first super geomagnetic storm of solar cycle 24: “The St. Patrick’s day event (17 March 2015)”. *Earth, Planets and Space*, 68(1):151, December 2016. doi: 10.1186/s40623-016-0525-y.
- R. K. Burton, R. L. McPherron, and C. T. Russell. An empirical relationship between interplanetary conditions and Dst. *Journal of Geophysical Research (Space Physics)*, 80(31):4204, November 1975. doi: 10.1029/JA080i031p04204.
- H. Xie, N. Gopalswamy, O. C. St. Cyr, and S. Yashiro. Effects of solar wind dynamic pressure and preconditioning on large geomagnetic storms. *Geophysical Research Letters*, 35(6):L06S08, March 2008. doi: 10.1029/2007GL032298.
- INTERMAGNET. International real-time magnetic observatory network, 2021. URL <https://www.intermagnet.org/index-eng.php>.
- Henry Rishbeth and Owen K. Garriott. *Introduction to ionospheric physics*. 1969.
- D.J Knecht and . Shuman B.M. *Handboor of Geophysics and the Space Enviromnent, chaper 4, The Geomagnetic Field*. ed by A. S. Jursa, Boston: Air Force Geophysics Laborator, 1985.
- J. W. Gjerloev. The SuperMAG data processing technique. *Journal of Geophysical Research (Space Physics)*, 117(A9):A09213, September 2012. doi: 10.1029/2012JA017683.
- M. van de Kamp. Harmonic quiet-day curves as magnetometer baselines for ionospheric current analyses. *Geoscientific Instrumentation, Methods and Data Systems*, 2(2):289–304, November 2013. doi: 10.5194/gi-2-289-2013.
- C. Amory-Mazaudier, O. S. Bolaji, and V. Doumbia. On the historical origins of the CEJ, DP2, and Ddyn current systems and their roles in the predictions of ionospheric responses to geomagnetic storms at equatorial latitudes. *Journal of Geophysical Research (Space Physics)*, 122(7):7827–7833, July 2017. doi: 10.1002/2017JA024132.
- Shun Takahashi, Masahiko Takeda, and Yuji Yamada. Simulation of storm-time partial ring current system and the dawn-dusk asymmetry of geomagnetic variation. *Planetary and Space Science*, 39(6): 821–832, June 1991. doi: 10.1016/0032-0633(91)90087-Q.
- Yu. V. Yasyukevich, A. A. Mylnikova, and A. S. Polyakova. Estimating the total electron content absolute value from the GPS/GLONASS data. *Results in Physics*, 5:32–33, January 2015a. doi: 10.1016/j.rinp.2014.12.006.

- Yu. V. Yasyukevich, A. A. Mylnikova, V. E. Kunitsyn, and A. M. Padokhin. Influence of GPS/GLONASS differential code biases on the determination accuracy of the absolute total electron content in the ionosphere. *Geomagnetism and Aeronomy*, 55(6):763–769, November 2015b. doi: 10.1134/S001679321506016X.
- A. A. Mylnikova, Yu. V. Yasyukevich, V. E. Kunitsyn, and A. M. Padokhin. Variability of GPS/GLONASS differential code biases. *Results in Physics*, 5:9–10, January 2015. doi: 10.1016/j.rinp.2014.11.002.
- V. V. Kalegaev, K. Yu. Bakhmina, I. I. Alexeev, E. S. Belenkaya, Ya. I. Feldstein, and N. V. Ganushkina. Ring current asymmetry during a magnetic storm. *Geomagnetism and Aeronomy*, 48(6):747–758, December 2008. doi: 10.1134/S0016793208060078.
- Georgy Makarov. Geomagnetic indices ASY-H and SYM-H and their relation to interplanetary parameters. *Solar-Terrestrial Physics*, 8(4):36–43, December 2022. doi: 10.12737/stp-84202203.
- L. Thomas and F. H. Venables. The onset of the F-region disturbance at middle latitudes during magnetic storms. *Journal of Atmospheric and Terrestrial Physics*, 28(6-7):599–605, January 1966. doi: 10.1016/0021-9169(66)90073-0.
- T. J. Fuller-Rowell, M. V. Codrescu, R. J. Moffett, and S. Quegan. Response of the thermosphere and ionosphere to geomagnetic storms. *jgr*, 99(A3):3893–3914, March 1994. doi: 10.1029/93JA02015.
- Sebastián Solari, Marta Egüen, María José Polo, and Miguel A. Losada. Peaks over threshold (pot): A methodology for automatic threshold estimation using goodness of fit p-value. *Water Resources Research*, 53(4):2833–2849, 2017. doi: <https://doi.org/10.1002/2016WR019426>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2016WR019426>.
- James Pickands. Statistical Inference Using Extreme Order Statistics. *The Annals of Statistics*, 3(1):119–131, 1975. doi: 10.1214/aos/1176343003. URL <https://doi.org/10.1214/aos/1176343003>.
- J. R. M. Hosking. L-moments: Analysis and estimation of distributions using linear combinations of order statistics. *Journal of the Royal Statistical Society: Series B (Methodological)*, 52(1):105–124, 1990. ISSN 0035-9246. doi: 10.1111/j.2517-6161.1990.tb01775.x. URL <https://doi.org/10.1111/j.2517-6161.1990.tb01775.x>.